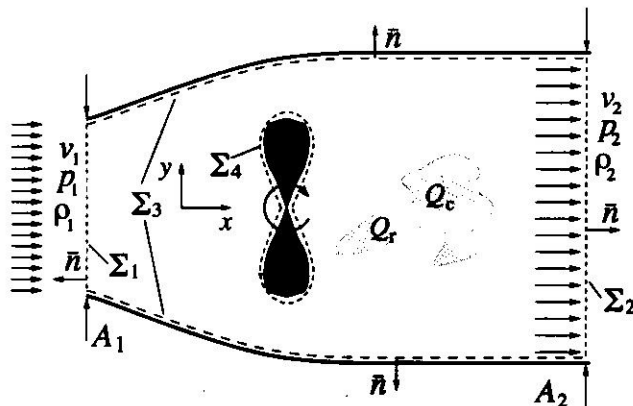


CL5

The entrance and exit sections of the turboreactor sketched in the figure have areas A_1 and A_2 . Besides the outer fixed frame, the machine has mobile parts in its interior that extract an amount of work W per unit time from the fluid stream. The combined effects of chemical reaction in the combustion chamber and heat transfer through the walls give a heat transfer per unit time Q to the fluid stream. The axial force on the turboreactor measured on the testbench is $\bar{F} = -F\bar{e}_x$. For given values of W , Q and F and assuming that at the entrance and exit sections the flow properties are uniform, obtain the values of the density ρ_2 , pressure p_2 and velocity v_2 in terms of their values ρ_1 , p_1 and v_1 at the entrance.

NO GRAVITY!



can: $\frac{d}{dt} \int_{V_c} \rho dV + \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} \rho (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0 \Rightarrow \rho_1 v_1 A_1 = \rho_2 v_2 A_2$

$\bar{v}_c = \bar{v} = 0$ $\bar{v}_c = \bar{v}_w$

$W = - \int_{\Sigma_{MP}} P \bar{v}_w \cdot \bar{n} + \int_{\Sigma_{MP}} \bar{v}_w \cdot \bar{e}^i \cdot \bar{n} d\sigma$

EXTENDED ON THE SOLID

MOMENTUM: $\frac{d}{dt} \int_{V_c} \rho \bar{v} dV + \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} \rho \bar{v} (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = - \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} (P - P_a) \bar{n} d\sigma + \int_{\Sigma_4} P \bar{n} d\sigma + \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} \bar{e}^i \cdot \bar{n} d\sigma$

$\bar{e}^i = 0$

X-COMPONENT: $\bar{e}_x \rho_1 v_1 A_1 [v_2 - v_1] = (P_1 - P_a) A_1 \bar{e}_x - (P_2 - P_a) A_2 \bar{e}_x + \int_{\Sigma_3} (P - P_a) \bar{n} d\sigma + \int_{\Sigma_4} \bar{e}^i \cdot \bar{n} d\sigma - \int_{\Sigma_4} P \bar{n} d\sigma + \int_{\Sigma_4} \bar{e}^i \cdot \bar{n} d\sigma$

$-F = F \bar{e}_x$

$F = \rho_1 v_1 A_1 (v_2 - v_1) - (P_1 - P_a) A_1 + (P_2 - P_a) A_2$

ENERGY: $\frac{d}{dt} \int_{V_c} \rho \left(e + \frac{v^2}{2} \right) dV + \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} \rho \left(e + \frac{v^2}{2} \right) (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = - \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} P \bar{v}_w \cdot \bar{n} d\sigma + \int_{\Sigma_4} \bar{v}_w \cdot \bar{e}^i \cdot \bar{n} d\sigma + \int_{\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4} k \nabla T \cdot \bar{n} d\sigma + \int_{V_c} (Q_c + Q_r) dV$

$\bar{v}_c = 0$ $\bar{v}_c = \bar{v}_w$ $\nabla T = 0$

$\rho_1 v_1 A_1 \left[\left(e_2 + \frac{v_2^2}{2} \right) - \left(e_1 + \frac{v_1^2}{2} \right) \right] = P_1 v_1 A_1 - P_2 v_2 A_2 - \int_{\Sigma_4} P \bar{v}_w \cdot \bar{n} d\sigma + \int_{\Sigma_4} \bar{v}_w \cdot \bar{e}^i \cdot \bar{n} d\sigma + \int_{\Sigma_3 + \Sigma_4} k \nabla T \cdot \bar{n} d\sigma + \int_{V_c} (Q_c + Q_r) dV$

$-W$ Q

$\rho_1 v_1 A_1 \left[\left(\frac{\gamma}{\gamma-1} \frac{P_2}{\rho_2} + \frac{v_2^2}{2} \right) - \left(\frac{\gamma}{\gamma-1} \frac{P_1}{\rho_1} + \frac{v_1^2}{2} \right) \right] = -W + Q$